

Electronics and Battery Placement Menu

Where Should the Hub, Battery, Switch, and Wires Go?

Graduate Expectation Tags

- **[Red: Critical Thinking & Problem Solving]** We compare balance, access, wiring, repair, and safety trade-offs.
- **[Green: Personal Development]** We make responsible choices that protect electronics and support safe testing.
- **[Blue: Effective Communication]** We explain our layout clearly in sketches, tables, and portfolio notes.
- **[Orange: Cultural Understanding & Civic Engagement]** We protect borrowed electronics and shared equipment.

Decision Being Made

Your team is deciding where to place the robot's electronics and battery. These parts are essential. If the hub, battery, switch, or wires are hard to reach or poorly mounted, the robot will be harder to test, repair, and improve.

Important parts include:

- Control Hub or Expansion Hub,
- battery,
- power switch,
- motor wires,
- servo wires,
- sensor wires if used,
- and USB or communication cables if needed.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Motor Placement Menu**
- **Chassis Physics and Space Planning**
- **Robotics Safety Norms**
- **Borrowed Equipment and Community Trust**
- your team's chassis sketch.

Why This Decision Matters

Electronics placement affects:

- whether wires can reach motors and servos,
- whether the battery can be changed quickly,
- whether the power switch is easy to reach,
- whether the robot is balanced,
- whether the hub lights and ports are visible,
- whether damaged parts can be replaced,
- whether future mechanisms can be added,
- and whether electronics are protected from contact.

Good electronics layout is not just about making the robot look clean. It is about making the robot reliable, repairable, and safe.

Required Layout Rule

The **power switch must be between the battery and the hub** and must be easy to reach. A teammate, mentor, or inspector should be able to find and use the switch quickly.

The battery should also be easy enough to remove and replace, because fresh batteries are needed often.

Menu Options

Option A: Hub Mounted Flat on the Floor of the Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: The hub is mounted flat on a low plate or frame surface inside the robot.

What it helps: This can keep the hub low and protected. It may lower the robot's center of mass and make the robot more stable.

Best for: Teams that want a low, protected electronics layout.

Ease of use: Easy to medium

Likelihood of success: Medium to high

Time to implement: Moderate

Possible trade-offs: Top ports may be easy to see, but bottom or side connectors may be harder to reach depending on the hub orientation. If mechanisms are built above it, access may become difficult.

Evidence we might watch: Can we plug and unplug wires easily? Can we see hub lights? Can we replace or inspect wires after mechanisms are added?

Option B: Hub Mounted Vertically on a Side Wall

Graduate Expectation Connection: [Blue: Effective Communication] and [Red: Critical Thinking & Problem Solving]

What it means: The hub is mounted upright on the side of the chassis or on a vertical plate.

What it helps: This can make ports, labels, wires, and lights easier to see. It may also keep the floor of the robot open for game pieces or mechanisms.

Best for: Teams that expect to add motors, servos, sensors, or wires during the course.

Ease of use: Medium

Likelihood of success: High if mounted securely

Time to implement: Moderate

Possible trade-offs: The hub may be more exposed to contact. It may raise some weight higher in the robot. It needs a strong mount so it does not shake loose.

Evidence we might watch: Can we reach the ports? Are wires organized? Is the hub protected enough during normal driving?

Option C: Hub Mounted High for Easy Access

Graduate Expectation Connection: [Green: Personal Development]

What it means: The hub is mounted where students can easily see and reach ports without removing other parts.

What it helps: This makes debugging, adding parts, checking wires, and reading lights easier.

Best for: Teams that expect frequent changes to wiring and code configuration.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Possible trade-offs: Mounting electronics higher can raise the center of mass. It may also expose the hub to contact or game pieces.

Evidence we might watch: Can we troubleshoot quickly? Does the robot remain stable during turns and contact?

Option D: Hub Hidden and Protected Inside the Frame

Graduate Expectation Connection: [Orange: Cultural Understanding & Civic Engagement]

What it means: The hub is placed inside the robot where frame pieces or plates protect it from contact.

What it helps: This can protect expensive borrowed electronics from bumps, game pieces, and robot contact.

Best for: Teams that prioritize protecting electronics and have already planned wire access carefully.

Ease of use: Hard

Likelihood of success: Medium

Time to implement: Slow

Possible trade-offs: Hidden electronics can become hard to repair, inspect, or rewire. If a wire comes loose, the team may need to remove other parts to fix it.

Evidence we might watch: Can we still reach every important port? Can we replace damaged parts? Can we see the status lights?

Option E: Battery Low and Centered

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The battery is mounted low in the robot, near the center of the chassis.

What it helps: The battery is heavy. Placing it low and near the center can improve balance and reduce tipping.

Best for: Teams that want stable driving and predictable turning.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Possible trade-offs: The battery may be harder to remove if it is buried inside the frame. The team must still be able to replace it quickly.

Evidence we might watch: Does the robot stay stable during turns? How long does it take to change the battery?

Option F: Battery Easy to Remove

Graduate Expectation Connection: [Green: Personal Development]

What it means: The battery is placed where students can quickly disconnect, remove, replace, and secure it.

What it helps: Battery changes happen often. Easy access saves time and reduces stress before testing or matches.

Best for: Teams that expect frequent testing and driving practice.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Fast to moderate

Possible trade-offs: The most accessible battery location may not be the best balance location. It may also take up space needed for mechanisms.

Evidence we might watch: Can we replace the battery quickly? Is the battery secure? Does it shift during driving?

Option G: Switch on the Outside Edge

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The power switch is mounted on an outside edge of the robot where it is easy to see and reach.

What it helps: This supports safe testing because the robot can be turned off quickly.

Best for: All teams. The switch should be easy to reach.

Ease of use: Easy

Likelihood of success: High

Time to implement: Fast

Possible trade-offs: The switch must be protected enough that it does not get hit accidentally, but it cannot be hidden.

Evidence we might watch: Can someone find and turn off the robot quickly? Is the switch protected from accidental bumps?

Option H: Electronics on One Side, Mechanisms on the Other

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The team organizes the robot so electronics and wires are mostly on one side, while mechanisms such as shooter, hopper, or trigger are on another side or area.

What it helps: This can make wiring cleaner and reduce conflicts between moving parts and electronics.

Best for: Teams that want organized wiring and clearer repair zones.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Possible trade-offs: Putting too much weight on one side can affect balance. Wires may need to travel farther to reach motors or servos.

Evidence we might watch: Does the robot pull or tip to one side? Are wires clear of moving parts? Can mechanisms move without hitting electronics?

Comparison Table

Option	Best Feature	Main Risk	Good For
Hub flat on floor	Low and protected	Harder access if covered	Stable compact layouts
Hub vertical on side	Easier port access	More exposed	Frequent wiring changes
Hub high for access	Easy troubleshooting	Higher center of mass	Debugging and additions
Hub hidden/protected	Protects electronics	Harder repair and inspection	Carefully planned layouts
Battery low/centered	Better balance	Harder battery changes	Stability
Battery easy to remove	Fast replacement	May affect balance/space	Frequent testing
Switch outside edge	Fast shutoff	Must avoid accidental hits	Safety and inspection
Electronics one side	Organized wiring	Side-heavy risk	Clean layout and repair zones

Layout Questions

Before finalizing electronics and battery placement, answer:

- Can we reach every motor, servo, and sensor port?
- Can we see hub lights?
- Can we replace a damaged wire?
- Can we add another motor or servo later?
- Can we remove and replace the battery quickly?

- Is the battery secure?
- Is the battery low enough to help balance?
- Is the power switch between the battery and hub?
- Is the power switch easy to reach?
- Are wires protected from wheels, gears, flywheels, and moving mechanisms?
- Does the layout leave room for shooter, hopper, trigger, and ball/artifact movement?

Electronics and Battery Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Hub orientation				
Hub location				
Battery location				
Switch location				
Wiring path				

Must / Should / Could

Level	Expectation
Must Do	Choose locations for the hub, battery, and switch and show them on the chassis sketch.
Should Do	Explain how the layout affects access, balance, wiring, safety, and future mechanisms.
Could Do	Compare two electronics layouts using battery-change time, wiring access, balance, or repair access.

SLD Connection

Main SLD Prompt

Should our electronics layout prioritize protection, easy access, balance, or open space for mechanisms?

Supporting Questions

- Is it better for the hub to be flat or vertical?
- Where can the battery go so the robot stays balanced but the battery is still easy to replace?
- How can the switch be easy to reach but protected from accidental hits?
- What parts will we need to add later?

- What happens if a wire comes loose during testing?
- How can we protect borrowed electronics while still making repairs possible?

Portfolio Deliverable

Add the following to your engineering portfolio:

- hub orientation and location,
- battery location,
- switch location,
- wiring path notes,
- labeled chassis sketch,
- explanation of the main trade-off,
- and evidence the team will watch during testing.

Reflection

Our hub placement choice is:

Our battery placement choice is:

Our switch placement choice is:

The main trade-off we accepted is:

The evidence we will watch during testing is:
